

flspcell_slow_1p08V_125C Library

Cell Groups
DFRBPQ

DFRBPQ



*flspcell_slow_1p08V_125C Cell Library: Process
flspcell_slow_1p08V_125C, Voltage 1.08, Temp 125.00*

Truth Table

INPUT			OUTPUT
CLK	D	RESET_B	Q
R	0	1	0
R	1	1	1
x	x	0	0
x	x	1	IQ

Footprint

Cell Name	Area
sg13g2_dfrbpq1arx_1	48.98880

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	CLK	D	RESET_B	Q
sg13g2_dfrbpq1arx_1	0.02125	0.01476	0.02575	0.30000

Leakage Information

Cell Name	Leakage(pW)		
	Min.	Avg	Max.
sg13g2_dfrbpq1arx_1	11453.80000	13270.90000	15648.40000

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_dfrbpq1arx_1	CLK->Q (RR)	0.01860	0.00100	0.58542	0.32940	0.06480	0.99555	2.50740	0.30000	2.35625

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_dfrbpq1arx_1	CLK->Q (RF)	0.01860	0.00100	0.64797	0.32940	0.06480	1.05789	2.50740	0.30000	2.23283
	RESET_B->Q (FF)	0.01860	0.00100	0.76362	0.32940	0.06480	1.20516	2.50740	0.30000	2.60179

Constraint Information

Constraints(ns) for CLK rising :

Cell Name	Timing Check	Ref Pin(trans)	Constraint(ns)								
			Input Slew(ns)	Ref Slew(ns)	First	Input Slew(ns)	Ref Slew(ns)	Mid	Input Slew(ns)	Ref Slew(ns)	Last
sg13g2_dfrbpq1arx_1	min_pulse_width	CLK ()	0.01860	0.00000	0.18875	1.26300	0.00000	2.08496	2.50740	0.00000	4.13818

Constraints(ns) for CLK falling :

Cell Name	Timing Check	Ref Pin(trans)	Constraint(ns)								
			Input Slew(ns)	Ref Slew(ns)	First	Input Slew(ns)	Ref Slew(ns)	Mid	Input Slew(ns)	Ref Slew(ns)	Last
sg13g2_dfrbpq1arx_1	min_pulse_width	CLK ()	0.01860	0.00000	0.21118	1.26300	0.00000	2.08496	2.50740	0.00000	4.13818

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Constraint(ns)								
			Input Slew(ns)	Ref Slew(ns)	First	Input Slew(ns)	Ref Slew(ns)	Mid	Input Slew(ns)	Ref Slew(ns)	Last
sg13g2_dfrbpq1arx_1	hold	CLK (R)	0.01860	0.01860	-0.08314	1.26300	1.26300	-0.28333	2.50740	2.50740	-0.36894
	setup	CLK (R)	0.01860	0.01860	0.73845	1.26300	1.26300	0.87967	2.50740	2.50740	0.95630

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Constraint(ns)								
			Input Slew(ns)	Ref Slew(ns)	First	Input Slew(ns)	Ref Slew(ns)	Mid	Input Slew(ns)	Ref Slew(ns)	Last
sg13g2_dfrbpq1arx_1	hold	CLK (R)	0.01860	0.01860	-0.03423	1.26300	1.26300	-0.18619	2.50740	2.50740	-0.28040
	setup	CLK (R)	0.01860	0.01860	0.79224	1.26300	1.26300	0.89316	2.50740	2.50740	0.97401

Constraints(ns) for RESET_B rising :

Cell Name	Timing Check	Ref Pin(trans)	Constraint(ns)								
			Input Slew(ns)	Ref Slew(ns)	First	Input Slew(ns)	Ref Slew(ns)	Mid	Input Slew(ns)	Ref Slew(ns)	Last
sg13g2_dfrbpq1arx_1	recovery	CLK (R)	0.01860	0.01860	0.21029	1.26300	1.26300	0.42095	2.50740	2.50740	0.56374
	removal	CLK (R)	0.01860	0.01860	-0.18094	1.26300	1.26300	-0.38856	2.50740	2.50740	-0.52537

Constraints(ns) for RESET_B falling :

Cell Name	Timing Check	Ref Pin(trans)	Constraint(ns)								
			Input Slew(ns)	Ref Slew(ns)	First	Input Slew(ns)	Ref Slew(ns)	Mid	Input Slew(ns)	Ref Slew(ns)	Last
sg13g2_dfrbpq1arx_1	min_pulse_width	RESET_B ()	0.01860	0.00000	0.16953	1.26300	0.00000	2.08496	2.50740	0.00000	4.13818

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_dfrbpq1arx_1	CLK	0.01860	0.00100	0.10948	0.32940	0.06480	0.10750	2.50740	0.30000	0.11527

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_dfrbpq1arx_1	CLK	0.01860	0.00100	0.12450	0.32940	0.06480	0.12425	2.50740	0.30000	0.13571
	RESET_B	0.01860	0.00100	0.11519	0.32940	0.06480	0.11486	2.50740	0.30000	0.11625

Passive power(pJ) for CLK rising :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	0.01860	0.02801	0.32940	0.02676	2.50740	0.03587

Passive power(pJ) for CLK falling :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	0.01860	0.04769	0.32940	0.04646	2.50740	0.05496

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)					
		Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	(D * RESET_B * Q)	0.01860	0.01998	0.32940	0.01866	2.50740	0.02797
	(D * !RESET_B * * !Q)	0.01860	0.02801	0.32940	0.02676	2.50740	0.03587
	(!D * RESET_B * !Q)	0.01860	0.02940	0.32940	0.02814	2.50740	0.03730
	(!D * !RESET_B * * !Q)	0.01860	0.03311	0.32940	0.03186	2.50740	0.04097

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)					
		Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	(D * RESET_B * Q)	0.01860	0.04769	0.32940	0.04646	2.50740	0.05496
	(D * RESET_B * !Q)	0.01860	0.05070	0.32940	0.04955	2.50740	0.05804
	(D * !RESET_B * * !Q)	0.01860	0.02472	0.32940	0.02386	2.50740	0.03253
	(!D * RESET_B * Q)	0.01860	0.08614	0.32940	0.11662	2.50740	0.12433
	(!D * RESET_B * !Q)	0.01860	0.02159	0.32940	0.02070	2.50740	0.02940
	(!D * !RESET_B * * !Q)	0.01860	0.01921	0.32940	0.01828	2.50740	0.02699

Passive power(pJ) for D rising :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	0.01860	0.04355	0.32940	0.04283	2.50740	0.04458

Passive power(pJ) for D falling :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	0.01860	0.04336	0.32940	0.04243	2.50740	0.04545

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)					
		Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	CLK	0.01860	0.04200	0.32940	0.04114	2.50740	0.04405
	(!CLK * RESET_B)	0.01860	0.07129	0.32940	0.07041	2.50740	0.07324
	(!CLK * !RESET_B)	0.01860	0.04355	0.32940	0.04283	2.50740	0.04458

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)					
		Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	CLK	0.01860	0.04336	0.32940	0.04243	2.50740	0.04545
	(!CLK * RESET_B)	0.01860	0.05702	0.32940	0.05604	2.50740	0.05916
	(!CLK * !RESET_B)	0.01860	0.03240	0.32940	0.03161	2.50740	0.03352

Passive power(pJ) for RESET_B rising :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	0.01860	0.00867	0.32940	0.00831	2.50740	0.01045

Passive power(pJ) for RESET_B falling :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	0.01860	0.02174	0.32940	0.01983	2.50740	0.02301

Passive power(pJ) for RESET_B rising (conditional):

Cell Name	When	Power(pJ)					
		Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	(CLK * D * !Q)	0.01860	0.00867	0.32940	0.00831	2.50740	0.01045
	(CLK * !D * !Q)	0.01860	0.00436	0.32940	0.00440	2.50740	0.00442
	(!CLK * D * !Q)	0.01860	0.03655	0.32940	0.03551	2.50740	0.03824
	(!CLK * !D * !Q)	0.01860	0.00685	0.32940	0.00688	2.50740	0.00689

Passive power(pJ) for RESET_B falling (conditional):

Cell Name	When	Power(pJ)					
		Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	(CLK * D * !Q)	0.01860	0.13799	0.32940	0.13557	2.50740	0.14084
	(CLK * !D * !Q)	0.01860	-0.00279	0.32940	-0.00335	2.50740	-0.00355
	(!CLK * D * !Q)	0.01860	0.02174	0.32940	0.01983	2.50740	0.02301
	(!CLK * !D * !Q)	0.01860	-0.00566	0.32940	-0.00613	2.50740	-0.00629